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One Recipe, Five Clouds, One Query

Portable GPU Configuration with a Queryable Audit Layer

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THE PROBLEM

Running GPU clusters reliably is hard

Services × GPUs × operating systems × workloads: more combinations than one team can validate.

Version compatibility

Kernels, drivers, runtimes, operators, and Kubernetes must align.

Tribal knowledge

Known-good settings live in runbooks and internal pipelines.

Expensive failures

One mismatch can disrupt a GPU fleet and take days to diagnose.

300+	configuration values in a fully tuned training cluster
16+	recipe components (training example)
5	component changes, 20 new values, and 2 value changes: training → inference
2 + 20	component + value changes, H100 → GB200 on GKE (TCPXO → RDMA networking)
4 + 13	component + value changes, EKS → GKE (GB200 training)

What is AICR?

AI Cluster Runtime (AICR): an open-source engine for reproducible GPU cluster configurations.



Optimized

Tuned for specific hardware, service, OS, and workload



Validation built in

Constraint checks and functional validation



Reproducible

Pinned release + same inputs → repeatable configuration



Composable

Layered overlays: one desired state, public or private cloud



Verifiable

Release provenance, SBOMs, and optional bundle signatures

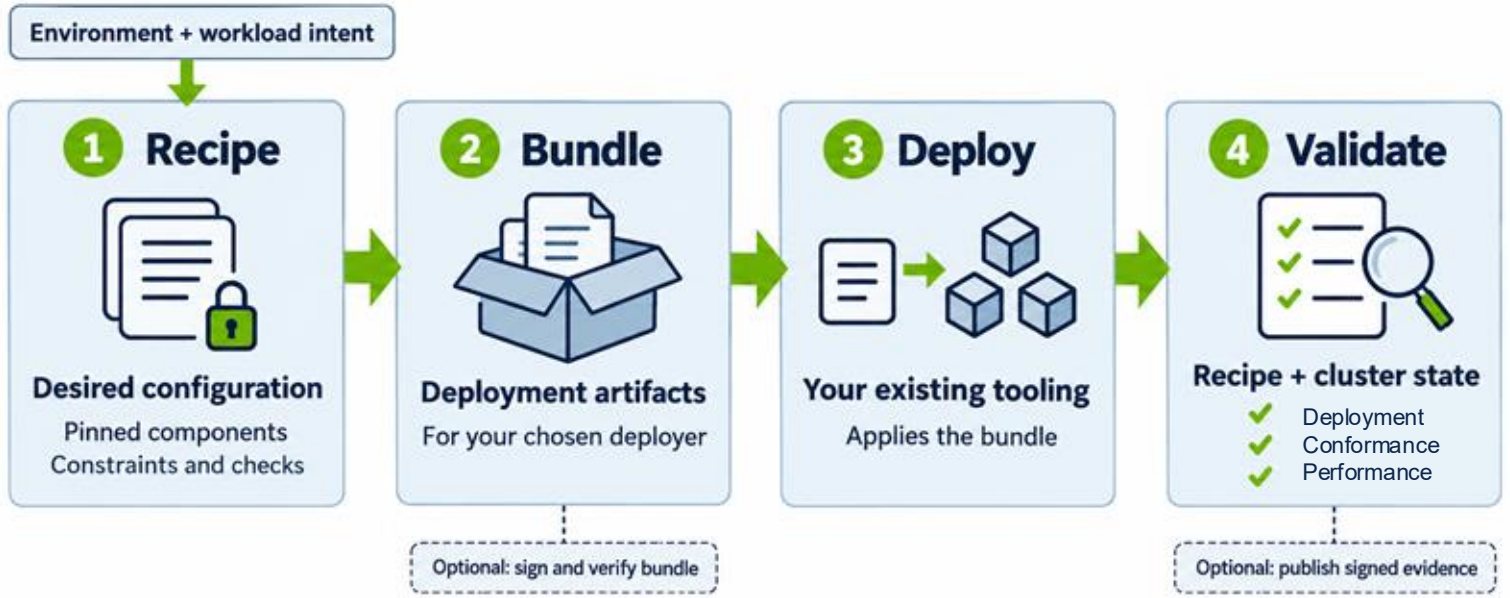


Fits your tooling

Helm, Argo CD, Flux, Helmfile, Terraform, Fleet + REST API

AICR workflow

Use criteria or aicr snapshot for environment facts. Choose intent and an optional platform.



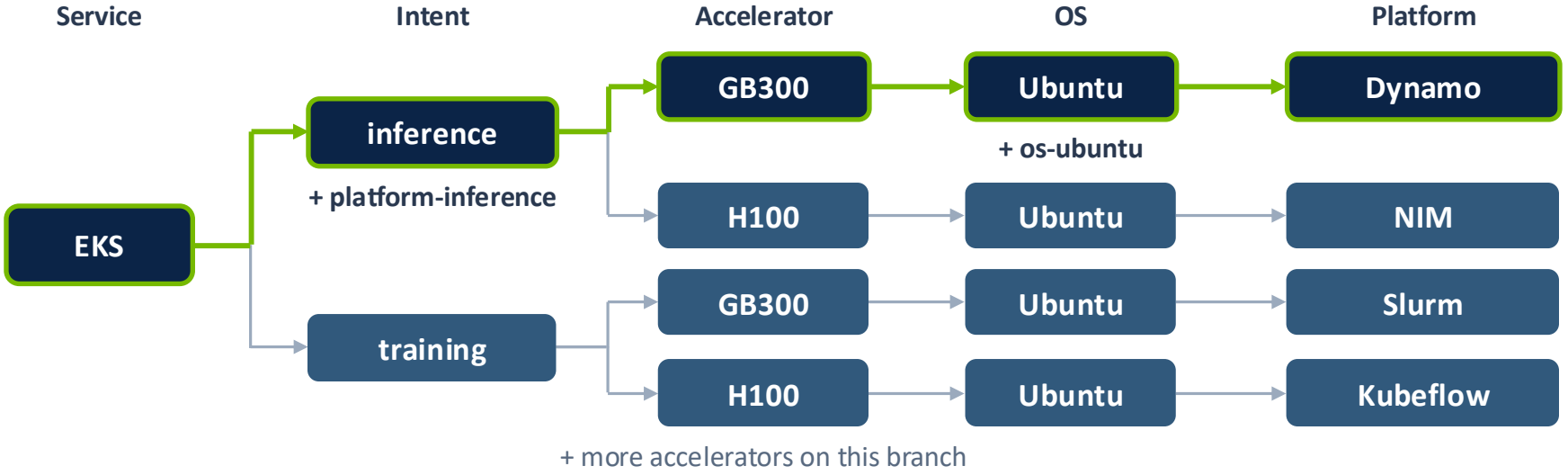
One recipe connects configuration, deployment, and validation.

RECIPE

Overlay hierarchy: the EKS branch

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Inherit a base, then specialize. The green path is the GB300 inference demo.



Demo overlay

`gb300-eks-ubuntu-inference-dynamo.yaml`

`spec.base → gb300-eks-ubuntu-inference`

Selected branches; shared base layers, matching overlays, and mixins also contribute.

From cluster facts to a recipe

Example: EKS · GB300 · Ubuntu · inference · Dynamo

Snapshot · observed

```yaml
fingerprint:
  service:
    value: eks
  accelerator:
    value: gb300
  os:
    value: ubuntu
```

User choices

inference · Dynamo

Compose shared knowledge

Base
↓
EKS
↓
EKS inference
↓
GB300
↓
Ubuntu
↓
Dynamo

+ 3 matching overlays · 2 mixins

Recipe · desired

```yaml
criteria:
  service: eks
  accelerator: gb300
  os: ubuntu
  intent: inference
  platform: dynamo
componentRefs:
- name: gpu-operator
  version: v26.7.2
```

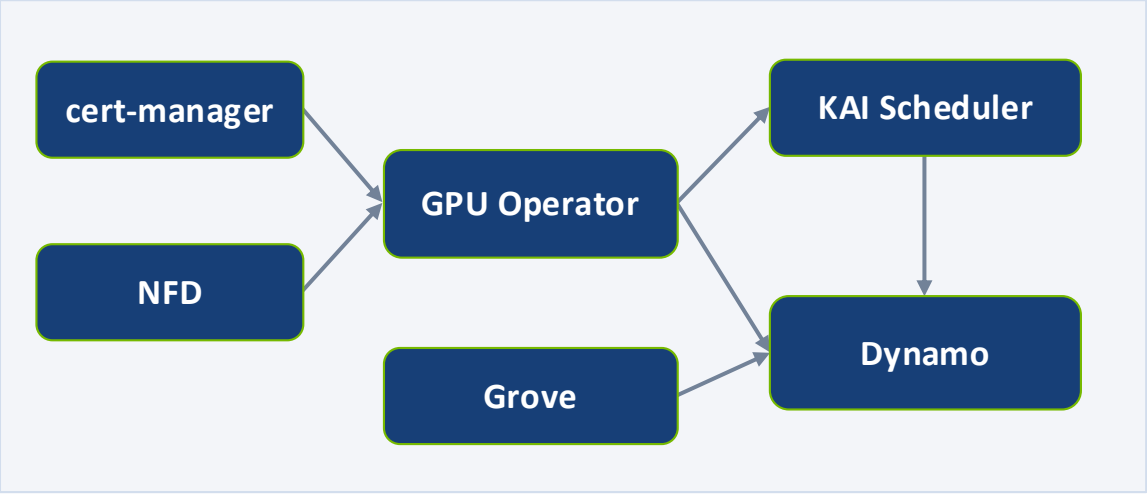
19 components

Shared layers compose one configuration for this workload.

From recipe to bundle

aicr bundle turns the recipe into deployable artifacts, in dependency order.

Recipe dependencies



Bundle · deployable

Helm example

```
bundle/
├─ recipe.yaml
├─ checksums.txt
├─ deploy.sh
├─ 015-gpu-operator/
│  └─ values.yaml
⋮
├─ 017-kai-scheduler/
├─ 018-dynamo-platform/
⋮
```

Dependency ordering · Checksums · Optional signing

Also renders for Argo CD · Flux · Helmfile · Terraform · Fleet

From bundle to deployment

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Verify the bundle, then deploy it with your tool, in the recipe's dependency order.

Deploy with the tool you already use

Helm	`./deploy.sh, in dependency order`
Helmfile	`helmfile apply`
Argo CD	`app-of-apps from Git or OCI`
Flux	`HelmReleases + kustomization.yaml`
Terraform	`terraform apply; depends_on keeps order`
Fleet	`gitrepo.yaml or helmops.yaml`

Deployment · ordered

Helm example

```
$ aicr verify ./bundle
Trust level: verified
$ ./deploy.sh
== Pre-flight checks
== Deploying components
├─ ✓ cert-manager
├─ ✓ gpu-operator
⋮
└─ ✓ prometheus-adapter
✓ 21/21 installed
```

Verify first · Dependency order in every deployer · Your tool owns the rollout

Next: validate the deployed cluster

Is the cluster ready and deployed?

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Check prerequisites first, then confirm the recipe's components are installed and healthy.

Readiness gate — Snapshot readings meet recipe prerequisites
Kubernetes · OS · kernel

Deployment

Are the recipe's components installed and healthy?

`operator-health`	GPU Operator pods running and healthy
`expected-resources`	Declared resources exist and are healthy
`gpu-operator-version`	Version meets the recipe constraint
`check-nvidia-smi`	Driver and GPU access on available GPU nodes

Next: Conformance →

A failed or unmeasurable prerequisite stops validation before any validator Job runs.
Then the phases run in order: deployment → conformance → performance.

Is the cluster AI-conformant?

CNCF AI Conformance checks exercise the capabilities AI workloads need.

CNCF AI Conformance

GPU access	`dra-support · secure-accelerator-access · driver-runtime-management`
Scheduling	`gang-scheduling`
Observability	`accelerator-metrics · ai-service-metrics`
Scaling	`pod-autoscaling · cluster-autoscaling`
Serving	`inference-gateway`
Operator health	`robust-controller · platform-health · gpu-operator-health`

12 checks on main; the recording ran 11

Next: Performance ➡

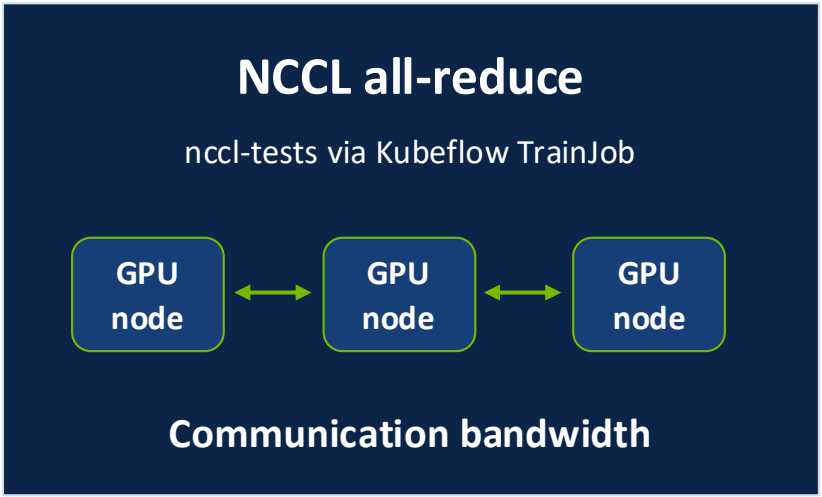
CI reports · CNCF submission evidence

NVKE 0.22.0 (NVIDIA Kubernetes Engine): CNCF AI Conformant for Kubernetes v1.36 · Validated with AICR recipes and tools

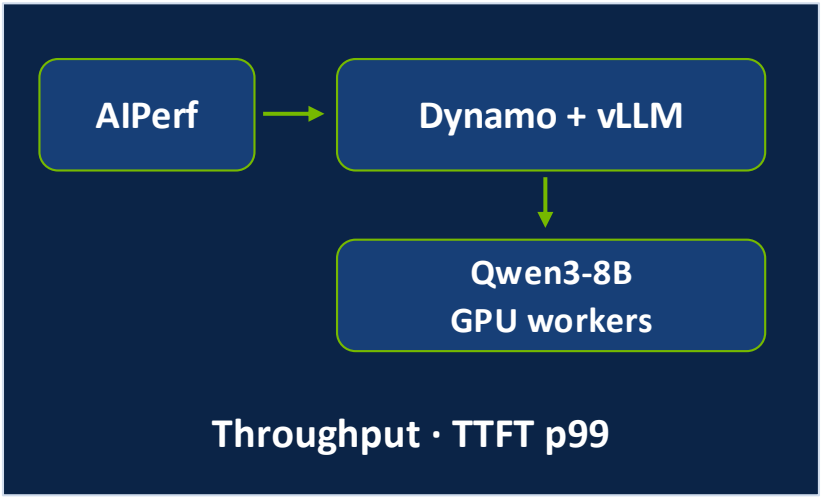
Does the cluster perform?

Run representative workloads against recipe-defined expectations.

Training infrastructure



Inference serving



Configurable inference: model · concurrency · routing strategy

DAY 2 · PREVIEW

Day 2: the AICR Operator

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Opt-in coordination for existing GPU clusters, using the same recipes and bundles.

Why an operator

Changing cluster facts
Re-resolve after cluster or AICR changes.

Current-revision readiness
Wait for the revision just applied to be ready.

Upgrade control
Approval when required; blocked moves stop.

Deployment evidence
Bind results to the rollout when enabled.

Who does what

AICR
Discover, resolve, render, assess, and validate.

AICR Operator
Plan, request approval, and order steps.

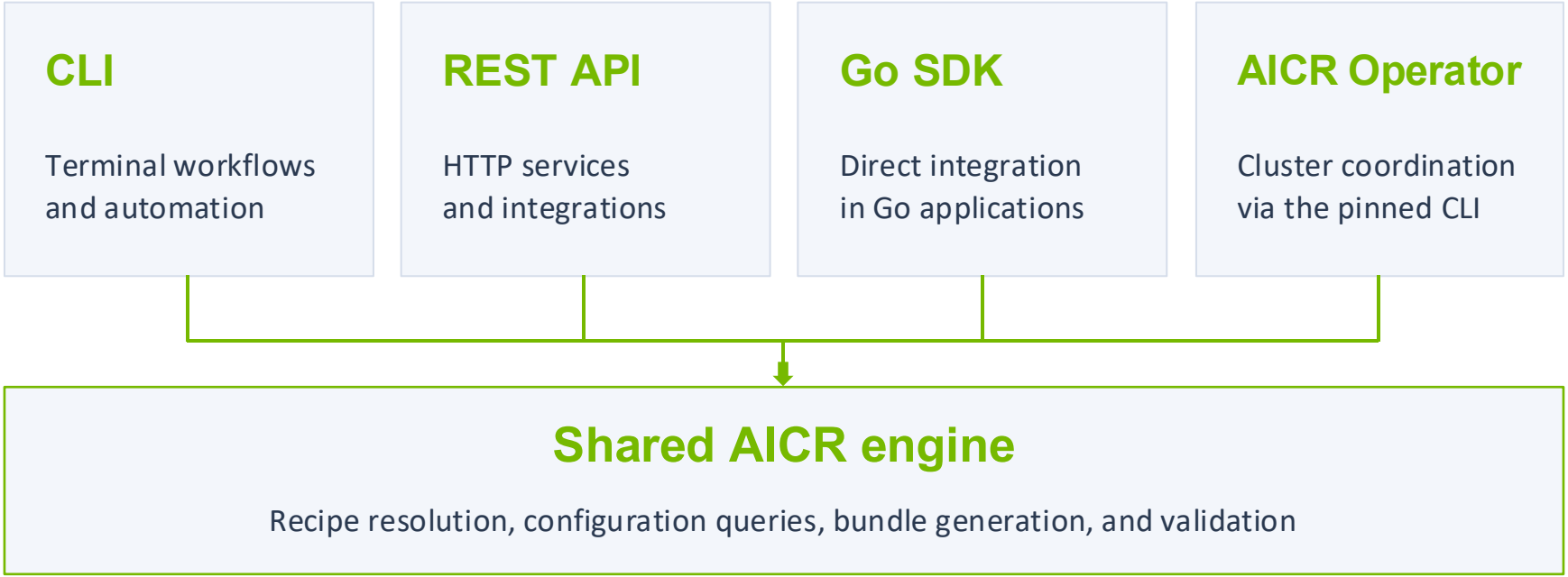
Argo CD
Install charts and reconcile their resources.

Component operators
Manage devices and component lifecycles.

ClusterRuntime → Plan → RuntimeRollout · assessed, gated, recorded

Four ways to use AICR

Terminal, HTTP, Go, and Kubernetes workflows share the same AICR engine.



DEMO · WHAT TO WATCH

Demo: end-to-end workflow

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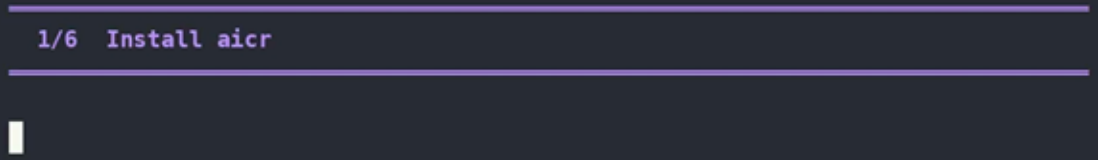
Recorded on a real EKS GB300 cluster. The deployment chapter replays dry-run output; validation was performed against the cluster.

1. **Capture and resolve**
   Snapshot the cluster; choose intent and platform; generate the recipe.

2. **Bundle and deploy**
   Generate and sign a Helm bundle, verify it, then replay deploy.sh dry-run output.

3. **Validate**
   Run deployment, conformance, and performance checks, and produce evidence.

Yuan · recording 2:13 · then live: Nemotron 3 Ultra deployment and chat



Play demo (2:13)

DEMO

Demo: Nemotron 3 Ultra, live

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Live: NVIDIA Nemotron 3 Ultra, served by Dynamo + vLLM on the AICR-built GB300 stack.



The model
Hybrid Mamba–Transformer MoE.

Total vs. active parameters
550B total; about 55B active per token.

Deployment footprint
~329 GiB NVFP4 weights · 256K served context.

NVIDIA Nemotron 3 Ultra model card · supports up to 1M tokens

Backup video of this demo (52 s)

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UP NEXT

Deep dive: the queryable audit layer



AUDIT LAYER

Four questions for an auditable cluster

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Inspect desired configuration, observed drift, upgrade guidance, and the evidence behind a result.

What should this cluster run?

```
aicr query
```

Resolve values; compare environments.

What changed since the baseline?

```
aicr diff
```

Field-by-field drift between two cluster snapshots.

Is this move safe?

```
aicr upgrade-check
```

Guidance from recorded transitions.

What can you verify?

```
Origin      cosign / gh attestation
Integrity   aicr verify
Results     aicr evidence verify
```

Signatures attribute results.
Validation checks cluster health.

AUDIT LAYER

What should this cluster run?

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Resolve configuration for your criteria, then select the value you need.

1. **Describe**
   EKS · GB300 · inference
2. **Resolve**
   Compose the recipe and merge component values.
3. **Select**
   GPU Operator → driver.version

Resolved answer

GPU Operator
`driver.version`
580.173.02
Recipe value ≠ installed driver

```
$ aicr query --service eks --accelerator gb300 --intent inference \
  --selector 'components.gpu-operator.values.driver.version'
```

Resolved configuration · CLI or REST API

AUDIT LAYER

One query, five clouds

GB300 inference examples: environment-specific components and GPU ownership.

EKS

Ubuntu

GPU Operator installs the driver

EFA networking
EBS CSI storage

GKE

COS · gke-default

GKE supplies driver and device plugin

Operator driver and plugin disabled

AKS

azure-managed

Node image supplies driver and toolkit

NVIDIA Network Operator

Also on managed clouds: OKE · LKE

Other environments: OpenShift · RKE2 · NVKE

Not every combination has a recipe.

What changed since the baseline?

Compare two observed snapshots to reveal added, removed, and modified fields.

Saved baseline
Reference cluster state

Fresh snapshot
Current observed state



Selected recorded changes

GPU Operator image
`v26.7.1 → v26.7.2`

NVSentinel labeler image
`v1.25.0 → v1.28.0`

Default StorageClass name
`Not recorded → ebs-csi-default-sc`

CI gate: --fail-on-drift

```
aicr diff --baseline baseline.yaml --target now.yaml --fail-on-drift
```

Validate the changed cluster against the recipe.

Is this move safe?

Assess the move against recorded transitions and deployer-specific steps.

1. **Compare**
   Current: recipe, bundle, or --from cluster
   Target: recipe or bundle
2. **Assess**
   Match transition records and deployer steps.
3. **Review**
   Read verdicts and required actions before applying.

GPU Operator
`v26.7.0 → v26.7.1` **safe**

NVSentinel
`v1.20.0 → v1.25.0` **blocked**

No recorded transition starts from NVSentinel v1.20.0.

```
$ aicr upgrade-check --from old.yaml --to new.yaml --deployer helm
```

Any verdict other than safe fails the default gate.

Evidence: collect, sign, and submit

Validation writes evidence, a signature attributes it, and a reviewed pointer publishes it.

1. **Collect**
   `--emit-attestation` writes a local evidence bundle of the results.

2. **Sign and push**
   `--push` signs it with Sigstore and publishes it to an OCI registry.

3. **Submit**
   A reviewed pointer and signer entry put it on validation.aicr.run.

Create, sign, and push to OCI:

```
$ aicr validate -r recipe.yaml \
  --emit-attestation ./out \
  --push ghcr.io/<org>/evidence
```

Only allowlisted signers count. Agreement and disagreement stay visible; results are never averaged.

Demo: inspect the audit trail

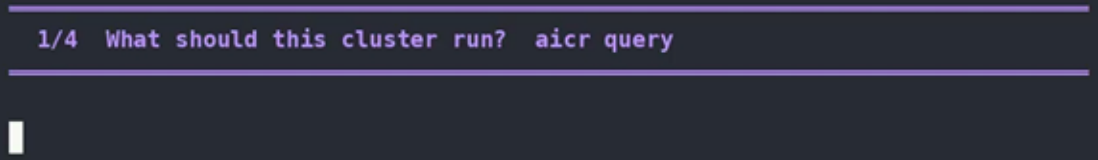
The same real cluster · four commands, four answers.

1. **`aicr query`**
   Read resolved values and compare environments.

2. **`aicr diff`**
   Compare observed snapshots with a saved baseline.

3. **`aicr upgrade-check`**
   Assess recorded transitions and deployer-specific steps.

4. **`aicr evidence verify`**
   Verify local evidence and a published signed pointer.



Play demo (1:34)

AUDIT LAYER

Live: the evidence portal

Live walkthrough: published results, their signers, and whether sources agree.

LIVE WALKTHROUGH

validation.aicr.run

Follow along on your laptop or phone



UP NEXT

Summary



ENVIRONMENTS

What's supported today

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Overlay criteria · AICR v1.2.0 + main (November 7, 2026)

13 Kubernetes services

Managed cloud	EKS · GKE · AKS · OKE · LKE · MK8s · CoreWeave CKS*
Distributions	OpenShift · RKE2* · k0s* · NVIDIA NVKE*
Self-managed	Bare metal · kind (dev/test)

11 GPU accelerators

Rubin	VR200*
Blackwell	GB300 · GB200 · B300 · B200 · RTX PRO 6000
Hopper	H200 · H100
Ampere / Ada	A100 · L40S · L40

5 Operating systems

Ubuntu · COS · Oracle Linux · Amazon Linux · SLES

Talos (mixin; not counted)

6 Deployer families

Helm · Helmfile · Argo CD · Flux · Terraform · Fleet

6 Workload platforms

Training	Kubeflow · Slurm · Kueue · Run:ai*†
Inference	Dynamo · NIM

** Preview. † On main; next release. Not every combination has a recipe. Argo CD has two formats: seven CLI formats across six deployer families.*

COMMUNITY

AICR community by the numbers

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GitHub snapshot: November 10, 2026 · 113 human PR/issue authors, excluding bots.

2,472
Pull requests 2,199 merged

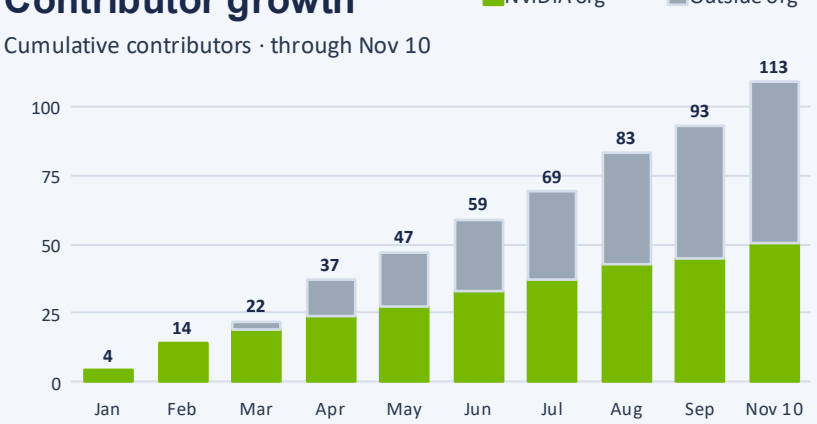
1,270
Issues 1,091 closed

113
Contributors PRs & issues

55%
Outside NVIDIA
GitHub org

Contributor growth

Cumulative contributors · through Nov 10



4 → 113 contributors since January

~190 merged PRs/month, excluding bots

25 organizations*

Identified from contributors' GitHub profiles

Akamai · Amazon · CDW · ConfigHub · Cornell University · DaoCloud · Dynamia AI
Elastic · Expanso · F5 · Gats.dev · Microsoft · Mirantis · NVIDIA
Oracle Cloud · PTC · Red Hat · RexCode Digital · Salesforce · Sidero Labs · Solo.io
SUSE · Turf · Upbound · Viettel Cloud

** Self-reported public profiles. Individual contributions do not imply organizational endorsement.*

GET INVOLVED

Use AICR, then help shape it

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The matrix is bigger than any one team. Use AICR, tell us what's missing, and help fill the gaps.

Use AICR and tell us what's missing

Report gaps and issues, and share feedback and suggestions.

Broaden coverage

- Validate existing recipes and submit signed evidence.
- Cover new platforms and GPUs: recipes, validation, and evidence.

Enhance tools and add features

File issues and help resolve them

Fix bugs and improve the documentation.



Explore AICR

RESOURCES

References and resources

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Everything shown today, and where to go next. Underlined items are links.

Get started

- Repo: **github.com/NVIDIA/aicr**
- Docs: **docs.nvidia.com/aicr**
- **End-to-end tutorial**
- `curl -sfL https://get.aicr.run | bash -s --`
- `brew tap NVIDIA/aicr && brew install aicr`

Validation and provenance

- Evidence portal: **validation.aicr.run**
- **CNCF Kubernetes AI Conformance**
- **Releases, SBOMs, and attestations**

NVIDIA Technical Blog

- **DSX OS: configure, certify, operate**
- **AICR v1.0: open, stable, verifiable**
- **Vera Rubin: AI-ready on day one**
- **Validated runtime for NVIDIA Dynamo**
- **NVCRE: validate GPU cluster readiness**
- **Layered, reproducible recipes (intro)**

Community

- **#aicr on Kubernetes Slack** (**slack.k8s.io**)
- **Roadmap**
- **Adopters**
- **Contributing guide**

FEEDBACK · VISIT US

Thank you!

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Come see us at the NVIDIA booth, and scan the code to rate this session.

Visit NVIDIA at booth 170



Booth 170

Hall 1, Alpine Overlook

Come by to learn more about AICR and the DSX OS and Kubernetes open-source projects.

Rate this session

One Recipe, Five Clouds, One Query: Portable GPU Configuration with a Queryable Audit Layer